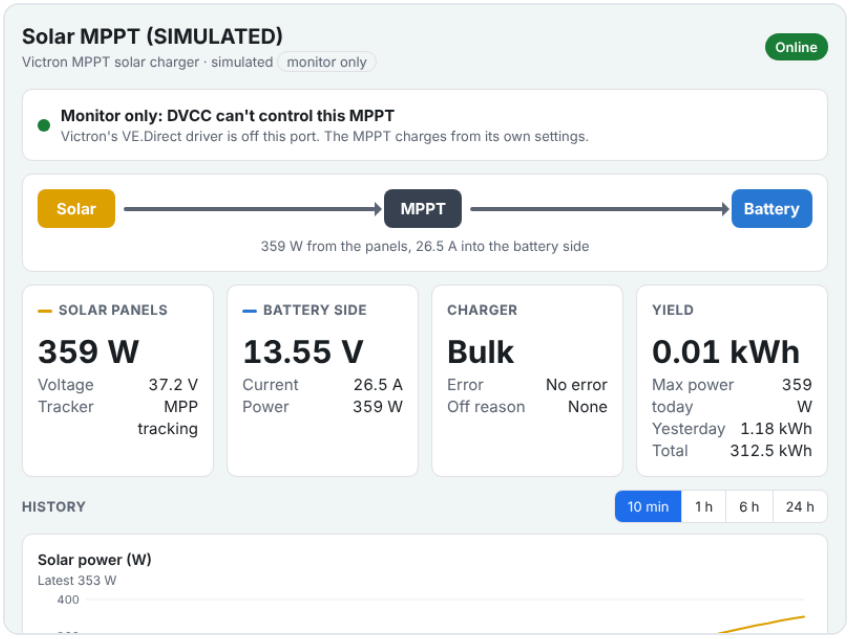
INSTALLATION AND USER GUIDE · MONITOR V1.0

Monitor a Victron MPPT on your Cerbo GX, without DVCC controlling it

Plug your MPPT solar charger into the Cerbo with a VE.Direct cable and install this monitor. You'll then see the MPPT's readings on the Cerbo screen, in the VRM app, and on a web page for your tablet. The MPPT keeps charging from **its own settings**: the Cerbo's DVCC can't change them.



The monitor's web page on a tablet (shown here with the built-in pretend MPPT).

You'll be able to see

- Panel voltage and power, and the battery-side voltage, current and power
- Charge state (Bulk, Absorption, Float…) and any error, in plain words
- Yield today, yesterday and in total, and the peak power of the day
- Charts of the last 24 hours on your tablet

Good to know

- It takes about 30 minutes
- It **only watches**: it opens the cable read-only and never sends the MPPT anything
- You don't need to type any commands. The **Cerbo Setup** helper does it for you
- Not yet tried on a real MPPT and Cerbo: step 6 shows how to check it on yours

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Why this is needed

When you plug a Victron MPPT into a Cerbo GX, the Cerbo's own driver registers it as a **solar charger**. If **DVCC** (Distributed Voltage and Current Control) is on, the Cerbo then takes over the MPPT's charging. It sends the MPPT a charge voltage and current, and the MPPT shows **"External control"**. With a managed battery on CAN, like a Lithionics, the Cerbo switches DVCC on by itself and may lock it. Victron has no setting that leaves one MPPT out of DVCC.

In a Grand Design LTV1 the MPPT charges the **12 V AGM chassis batteries** from the rear 12 V bus bar; once they're full, the Nomadic D1 boosts the surplus to the 51 V Lithionics. DVCC assumes every solar charger feeds the battery its BMS manages, so it would apply the Lithionics' limits to an MPPT on a different battery.

How the monitor keeps DVCC away

- **It stops Victron's driver on that one port.** It asks Venus OS (the Cerbo's software) to stop its VE.Direct driver on the MPPT's port only, using Victron's own command for this. It checks every 10 seconds that the driver stays off, and shows the result on a **Control** tile. Your other VE.Direct ports are untouched.
- **It shows the MPPT as a "DC source", not a solar charger.** In Victron's own code, DVCC only controls solar chargers, inverter/chargers, alternators and DC generators. It never controls a DC source.
- **It opens the cable read-only.** The operating system itself refuses any attempt to send the MPPT data.

The one gap: the first seconds after each Cerbo restart

Victron's driver starts slightly before the monitor when the Cerbo boots. In that short window DVCC could send the MPPT a limit. The monitor then stops the driver, and the MPPT goes back to its own settings. Step 6 checks this on your system. The optional one-way cable, near the end of this guide, closes the gap completely.

What changes for you

Before	After installing the monitor
The Cerbo (DVCC) decides the MPPT's charge voltage and current	The MPPT charges from its own settings, which you set in VictronConnect (step 2)
Shown as a solar charger on the Cerbo and in VRM's solar widgets	Shown as **"Solar MPPT"**: cards in the switch pane, a DC source on the Overview, VRM's Switches panel, and its own web page
Settings could be changed through the Cerbo or VRM	Change settings with **VictronConnect over Bluetooth**, next to the MPPT. The Cerbo can't change the MPPT any more, on purpose

1 Get what you need

Item	Details
MPPT solar charger	A Victron BlueSolar or SmartSolar MPPT with a VE.Direct port
Cerbo GX	Running **Venus OS v3.80 or newer** (Settings › General › Firmware). Older versions don't show the switch-pane cards
VE.Direct cable	Victron's straight (part no. ASS030530xxx) or right-angle (ASS030531xxx) cable, long enough to reach. **10 m (33 ft) at most**, and it can't be extended. A VE.Direct-to-USB cable into a Cerbo USB port works too
VictronConnect app	On your phone or tablet, to set up the MPPT over Bluetooth (step 2)
A Windows PC	On the same network as the Cerbo, for the **Cerbo Setup** helper. No PC? Use the USB-stick install in step 5B
The monitor	From GitHub: `github.com/CKOCI13/Victron-Cerbo-monitor-a-MPPT-without-DVCC`. Click the green **Code** button, then **Download ZIP**, and unzip it on the PC

The MPPT has only one VE.Direct port, so you can't also plug in an MPPT Control display. The tablet takes its place.

2 Set up the MPPT itself (do this first)

> **Important**
>
> Once DVCC is out of the picture, **the MPPT's own settings decide how it charges**. AGM batteries have no BMS to protect them, so set the MPPT up carefully before you install the monitor.

2.1 Open **VictronConnect** on your phone or tablet, stand near the MPPT, and connect to it over Bluetooth. Then open the settings (the gear icon).

2.2 **Battery voltage:** switch off auto-detect and set it to **12 V**.

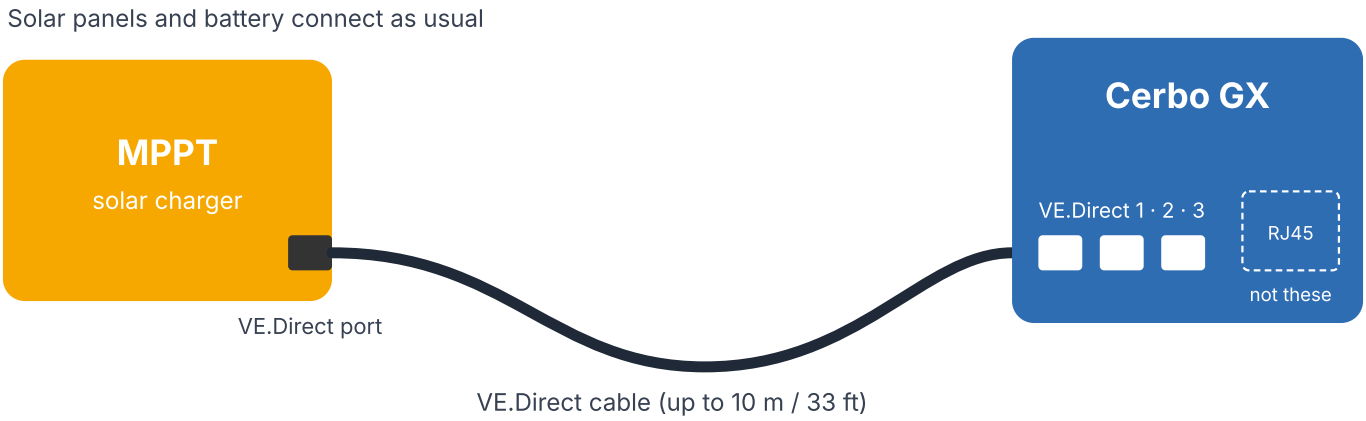
2.3 **Charge voltages:** use the AGM maker's absorption and float figures (or VictronConnect's **AGM** preset), and a **maximum charge current** the batteries' maker allows.

2.4 **Temperature compensation:** leave it **on** (about −16 mV/°C at 12 V). The MPPT uses its own internal temperature unless a Victron **Smart Battery Sense** on the batteries (under the driver's seat) sends it theirs over Bluetooth.

2.5 Write down the MPPT's **serial number** (VictronConnect › Product info, or the label on the MPPT; it looks like `HQ2245ABCDE`). There's a box for it on the last page.

2.6 **Match the D1:** its boost start voltage (12 V side) should sit **at or below the MPPT's float**, or no solar reaches the house battery, but **not far below absorption**, or the AGMs may never fully charge.

3 Plug in the cable



> **Before you plug in: shared ground**
>
> A VE.Direct cable (the one-way cable too) carries a **ground wire** between the MPPT and the Cerbo. The MPPT's negative is the 12 V side, which returns through the Ford chassis grounds. If the Cerbo is powered from the house-battery side, confirm that the 51 V (48 V nominal) negative and the chassis negative are joined (for example through the D1). If they aren't, ask Victron before connecting.

3.1 Plug one end into the MPPT's **VE.Direct** port and the other into **any** of the Cerbo's three **VE.Direct** ports. They're small 4-pin sockets. They are **not** the RJ45 sockets marked VE.Can or VE.Bus.

3.2 For the next few minutes the Cerbo shows the MPPT as a solar charger, and DVCC may take control of it. That's expected: it's how the monitor finds the MPPT. It ends when you install the monitor in step 5.

4 Set up the Cerbo

On the Cerbo's screen, or in the Remote Console on your tablet:

4.1 **Check the firmware:** Settings › General › Firmware. It must be **v3.80 or newer**. If it isn't, update it there first.

4.2 **Allow the PC to connect** (only for step 5A):

- Settings › General › **Access level** → **Superuser**. Set a **root password** when asked, and write it down
- Settings › General › **SSH on LAN** → **On**

4.3 **Find the Cerbo's IP address:** Settings › Connectivity › Ethernet (or Wi-Fi) › IP address, for example `192.168.1.50`. Write it on the last page.

4.4 **Modification checks:** Settings › General › Modification checks. Modifications must be **enabled** (that's the default). When they're off, the monitor doesn't start after a restart.

5 Install the monitor

Choose **A** (from a Windows PC, over the network) or **B** (from a USB stick, no network needed).

5A · With the Cerbo Setup helper on a Windows PC

A1 In the unzipped folder, double-click **Cerbo Setup.bat**. Type the Cerbo's IP address when asked and press Enter. It's remembered for next time.

A2 Type **2** and press Enter: **Show the VE.Direct ports**. When asked for a password, type the Cerbo's root password. Nothing appears while you type; that's normal. You should see your MPPT:

```
VE.Direct ports (nothing is ever sent to any device):
  ttyS6    built-in VE.Direct
           Victron's driver sees: SmartSolar MPPT 150/35 HQ2245ABCDE (solarcharger)  <- DVCC
can control this
           open by: vedirect-dbus (pid 1234)
config.ini has port = auto: found SmartSolar MPPT 150/35 HQ2245ABCDE on ttyS6.
```

Write down the port name (here `ttyS6`). This step only looks: it changes nothing.

A3 **Only if you have more than one MPPT on VE.Direct:** type **1** to open the settings, and after `serial_number =` type the serial number of the MPPT to monitor. Save and close Notepad. The other MPPTs stay with Victron's driver. With one MPPT, skip this.

A4 Type **4**: **Install or update the monitor**. After about 15 seconds you should see `up (pid …)` and, in the log lines, `Control status: monitor_only`.

5B · From a USB stick or SD card

B1 On the GitHub page, open **Releases** and download **venus-data-mppt-monitor.tar.gz**. (No release listed yet? On a Windows PC, run Cerbo Setup.bat, press Enter at the IP-address question, and choose **U**: it makes the same file.) On a Mac, download with Chrome or Firefox: Safari can unpack it into a .tar file, which the Cerbo ignores.

B2 Copy the file to the **top folder** of a USB stick or SD card (FAT32 is the safe choice), not inside another folder. Only one `venus-data` file on the stick, and its name must not have `(1)` added.

B3 Put the stick in the Cerbo and restart it (Settings › General › Reboot).

B4 Wait about **2 minutes**, then take the stick out. The Cerbo renames the file on the stick to `…tar.gz.installed`, so it never installs twice.

Either way, the monitor now starts by itself whenever the Cerbo starts, including after firmware updates.

6 Check that it works

Do this once after installing, in daylight if you can. None of these steps can change the MPPT. Tick each box.

#	Where	What you should see	OK
1	Cerbo switch pane	A **Solar MPPT** card with **Control: Monitor only (DVCC can't reach it)**. A red **Victron driver active!** that stays for more than a minute is a problem: see Troubleshooting	☐
2	Cerbo › Settings › Devices	The MPPT is listed **once**, as **Solar MPPT**, and **not** as a solar charger	☐
3	Switch pane or web page	Battery voltage within about 0.2 V of the 12 V bus (D1 monitor or a meter), and panel power above zero in daylight	☐
4	VictronConnect on the MPPT (Bluetooth)	Wait 5 minutes after installing. The state is **Bulk, Absorption or Float**, **not "External control"**	☐
5	Tablet browser	`http://<cerbo-ip>:8090` opens and shows a **green** Control box	☐
6	Restart the Cerbo, wait 5 min	Repeat checks 1, 2 and 4. This checks the start-up gap: the MPPT must **not** be in External control	☐

> **If check 4 or 6 fails**
>
> If the MPPT stays in **External control** for more than a few minutes, uninstall the monitor (Cerbo Setup option 8, then restart the Cerbo) and unplug the VE.Direct cable. Then report it on the GitHub page under **Issues**, with the monitor's log (Cerbo Setup option 5).

7 Use it

On the Cerbo screen (or Remote Console)

Open the **switch pane**. Three cards appear; everything on them is read-only:

Card	Shows
Solar MPPT	**Control** (green: monitor only; red: Victron's driver is active), **State** (Off, Bulk, Absorption, Float, Fault...), **Error** (None, or the error number and its meaning), **Yield today**, **Max power today**. Hidden at first: Yield yesterday, Yield total, Tracker
Solar MPPT solar	Panel voltage and power
Solar MPPT battery	Battery-side voltage, charge current and power

To show a hidden item, rename one or move it to another card, open Settings › Devices › **Solar MPPT**. The MPPT also appears on the **Overview** as a DC source box, and in **VRM** (portal and app) under **Switches**.

On your Android tablet

7.1 Open Chrome and go to `http://<cerbo-ip>:8090`. The tablet must be on the same network as the Cerbo.

7.2 Make it an icon: Chrome menu (⋮) › **Add to Home screen**.

7.3 The page shows the Control status, energy flow, panel and battery values, charge state and errors, yields, and charts. Tap **10 min / 1 h / 6 h / 24 h** to change the chart range, and touch a chart to read the values at that moment.

The charts start empty after a restart: the history is kept in memory for 24 hours. For longer history, use VRM. To require a password for the page, set `password =` in the settings (Cerbo Setup option 1, then 4).

Changing the MPPT's settings

Use **VictronConnect over Bluetooth**, next to the MPPT, exactly as before. The Cerbo, Remote Console and VRM can't change the MPPT while the monitor runs. That's what keeps DVCC out. (A BlueSolar without Bluetooth: unplug the Cerbo cable, connect a laptop with a VE.Direct-to-USB cable, make the change, then plug the Cerbo cable back in.)

Error numbers

The Error item shows the MPPT's own error number with its meaning, for example **Error 33: PV over-voltage**. The MPPT solar charger manual, chapter "Error codes", explains what to do for each one.

Optional: a one-way cable

For a guarantee in hardware, so that nothing on the Cerbo can ever send the MPPT anything (even in the start-up seconds), use a cable that only carries data **from** the MPPT **to** the Cerbo.



1. Use a short **4-pin JST-PH extension lead** between the VE.Direct cable and the MPPT, so the Victron cable stays intact.
2. At the MPPT end, cut and insulate the wire on **pin 2 only** (pin numbers from Victron's VE.Direct protocol document).
3. In the settings (Cerbo Setup option 1), set `port =` to the port name from step 5A (for example `ttyS6`) instead of `auto`. Victron's driver may not recognise a one-way MPPT, so `auto` might never find it. Then reinstall (option 4).
4. Check with option **3** (Watch live readings) that readings still arrive. If they stop, the wrong wire was cut: pin 3 carries the readings. Cutting either data wire can't damage anything.

The temperature probe

The Cerbo's temperature inputs are separate from this monitor. Use Victron's **ASS000001000** temperature sensor on one of the Cerbo's four temperature inputs: **red wire to the top terminal, black to the bottom**. Then switch the input on under Settings › Integrations › Tank and Temperature Sensors. It shows in the Cerbo's temperature unit, so in **°F** if the Cerbo is set to Fahrenheit. The MPPT doesn't use this probe: the Cerbo would only pass it on through DVCC, and only to solar chargers, which this MPPT no longer is.

Living alongside the Nomadic D1 monitor

Both run on the same Cerbo without touching each other: different connections (CAN for the D1, VE.Direct here), folders, services, device instances (70 and 71) and web pages (port 8088 for the D1, 8090 here).

Troubleshooting

What you see	What to do
No Solar MPPT cards at all	Check the firmware is v3.80 or newer (step 4.1). Look at the log (Cerbo Setup option 5). After a USB install, check Modification checks (step 4.4)
Control: **Waiting for the MPPT**	The monitor hasn't found the MPPT. Check the cable is in a **VE.Direct** socket at both ends. Run option 2: if the MPPT isn't listed, restart the Cerbo so Victron's driver can identify it, then wait a minute
Log says `2 MPPTs found … set serial_number`	Set `serial_number` to the one to monitor (step 5A, A3), then reinstall (option 4)
Control: **Victron driver active!** for more than a minute	Victron's driver keeps coming back. Send the log (option 5) with an issue on GitHub. Meanwhile, uninstall (option 8) and restart the Cerbo
State: **No communication**	No readings for 5 seconds. Check the cable. A cable longer than 10 m, or one that has been extended, won't work reliably
Web page doesn't open	Use `http://`, not https, and include `:8090`. The tablet must be on the same network as the Cerbo
Values look frozen	The Last reading line under Details on the web page shows how long ago data arrived. Over 5 seconds means no data: check the cable

Removing the monitor

Run Cerbo Setup, choose **8** and type **YES**. Then **restart the Cerbo**. Victron's driver takes the port back on the next start, and the MPPT is a normal solar charger again, which DVCC can control.

My system

Item	Write it here
Cerbo IP address	
MPPT model and serial number	
VE.Direct port (from option 2)	
Web page	http://_____:8090
MPPT absorption / float / max current	
D1 boost start voltage	
Installed on (date) / checks 1–6 done	

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